

Senior Frontend Engineer · Hiring Panel · Design-to-Code Assessment

CONFIDENTIAL

FIGMA DESIGN — COMPONENTS / CARDS / PROPERTYCARD



MOCK DATA

property.json — hand this to the candidate

JSON

```
{
  "id": "prop_abc123",
  "status": "active", // "active" | "pending" | "sold"
  "listing": {
    "price": 485000,
    "priceHistory": [
      { "date": "2024-01-15", "price": 500000, "event": "listed" },
      { "date": "2024-02-20", "price": 485000, "event": "reduced" }
    ],
    "address": {
      "street": "123 Maple Ave",
      "unit": null, // ← skip line if null
      "city": "Austin",
      "state": "TX",
      "zip": "78701"
    },
    "details": {
      "beds": 3,
      "baths": { "full": 2, "half": 1 },
      "sqft": 1820,
      "yearBuilt": 1998,
      "features": {
        "interior": ["hardwood floors", "granite counters", "open kitchen"],
        "exterior": ["2-car garage", "large backyard"],
        "amenities": null // ← don't crash on null
      }
    }
  },
  "agent": {
    "name": "Sarah Jones",
    "photo": null,
    "contact": {
      "phone": "512-555-1234",
      "email": "sarah.jones@realestate.com",
      "office": "Downtown Office",
      "name": "Sarah Jones"
    },
    "openHouses": [
      { "date": "2024-03-10", "time": "12:00-2:00" },
      { "date": "2024-03-17", "time": "12:00-2:00" }
    ],
    "media": {
      "photos": [
        { "url": "https://example.com/photo1.jpg" },
        { "url": "https://example.com/photo2.jpg" },
        { "url": "https://example.com/photo3.jpg" },
        { "url": "https://example.com/photo4.jpg" }
      ],
      "virtualTour": "https://example.com/virtual-tour"
    }
  }
},
```

INTERVIEW QUESTIONS

1

Implementation

45 min · Build the **PropertyCard** from the Figma spec

Using the JSON and Figma above, build a **PropertyCard** React component.

It should show:

- Primary photo + photo count badge

2

Architecture

30 min · Refactor for a shared design system

Your card works. Now it's going into a shared design system used by 4 teams.

Walk through and implement:

1. **Folder structure** — what's the public API surface?

- Price, status badge (color from design tokens)
- Address — skip unit if null
- Beds · baths · sqft · year built
- First 3 interior features only
- Price history with event styling
- Agent — initials avatar if photo is null

AREA	✔ PASS	❌ FAIL
Baths	full + half×0.5 → "2.5"	Hardcodes or ignores half
Null unit	Omits line entirely	Renders blank or "null"
amenities: null	Guards with ?. before map	Crashes on null
agent.photo: null	Initials "SC" avatar	Broken img tag
Status color	Maps to token/CSS class	Inline style hardcoded
Primary photo	find(isPrimary) ?? [0] ?? null	Always uses index 0

2. **Component extraction** — what becomes its own component?
3. **usePropertyCard hook** — what derived logic lives here?
4. **Utils** — what's a pure, testable function?
5. **Types** — define the full Property type
6. **Edge cases** — what does Q1 still not handle?

AREA	✔ PASS	❌ FAIL
index.ts	Only exports public API	Exports internals directly
Hook	All derived data in hook	Inline logic in JSX
Utils	formatCurrency, formatBaths pure fns	Logic mixed into components
photos: []	Placeholder + hide count	Not handled
price: 0	"Price not disclosed"	Shows \$0
status: "sold"	Red badge from token	No visual difference